

So far, our computer programs (R scripts) all ran sequentially.

Now let's learn about some **control structures**.

Ways to control the flow of your program.

Conditions

Conditions

Indicate something that must be TRUE in order for some code to be executed.

Humans

- "Make sure to drip your faucets when the temperature drops below 15 degrees Fahrenheit."
- "We cannot run a marathon unless we prepare and train regularly."
- "If you have a question, please raise your hand."

Computers

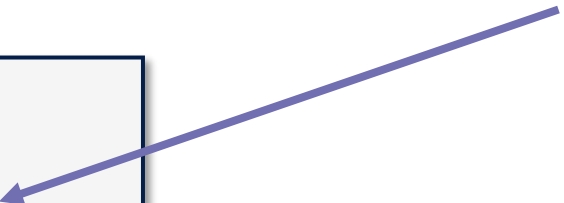
```
computer_programming <- "fun"  
if (computer_programming == "fun"){  
  print("That's right")  
}
```

Boolean

TRUE or FALSE.

This is **TRUE**.

```
computer_programming <- "fun"  
if (computer_programming == "fun"){  
  print("That's right")  
}
```



Exercise

Boolean Expressions

Take a look at the code below:

```
# Assign to an object called `x`, the result of `1 == 1`  
x <- 1 == 1  
  
# Assign to an object called `y`, the result of `2 < 1`  
y <- 2 < 1
```

What are the values of x and y?

- a. x = TRUE, y = TRUE
- b. x = TRUE, y = FALSE
- c. x = FALSE, y = TRUE
- d. x = FALSE, y = FALSE

Exercise

Boolean Expressions

Take a look at the code below:

```
# Assign to an object called `x`, the result of `1 == 1`  
x <- 1 == 1  
  
# Assign to an object called `y`, the result of `2 < 1`  
y <- 2 < 1
```

What are the values of x and y?

- a. x = TRUE, y = TRUE
- b. x = TRUE, y = FALSE**
- c. x = FALSE, y = TRUE
- d. x = FALSE, y = FALSE

If... Else If... Else...

Humans

If it is hot today, we should stay indoors.

Otherwise if it's too cold, we should also stay indoors.

Otherwise, we should go on a walk outside.

Computers

```
if (computer_programming == "fun"){  
    print("That's right")  
} else if (computer_programming == "boring"){  
    print("That's wrong")  
} else{  
    print("Interesting...")  
}
```


Demo

Boolean Values and
Expressions

In this demo, we will:

Evaluate Boolean expressions.

```
computer_programming <- "fun"  
if (computer_programming == "fun"){  
  print("That's right")  
}
```

Knowledge Check

Conditions

Reorder the following code snippets:

a)

```
} else{  
    print("We got some work to do!")  
}
```

b)

```
if (grade >= 90){  
    print("That's great")  
}
```

c)

```
} else if (grade >= 70){  
    print("You passed")  
}
```

Knowledge Check

Conditions

Reorder the following code snippets:

b)

```
if (grade >= 90){  
    print("That's great")  
}
```

c)

```
} else if (grade >= 70){  
    print("You passed")  
}
```

a)

```
} else{  
    print("We got some work to do!")  
}
```



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